

CS 660: Data Mining For Decision Making

Lecture 20 (Week 10)

Varun Dutt

School of Computing and Electrical Engineering
School of Humanities and Social Sciences
Indian Institute of Technology Mandi, India



Scaling the Heights

Discriminant Analysis

The purpose of Discriminant Analysis is to classify objects (people, customers, things, etc.) into one of two or more groups based on a set of features that describe the objects (e.g. gender, age, income, weight, preference score, etc.). In general, we assign an object to one of a number of predetermined groups based on observations made on the object.

Note that the groups are known or predetermined and do not have order (i.e. nominal scale). The classification problem gives several objects with a set features measured from those objects. What we are looking for is two things:

1. Which *set of features* can best determine group membership of the object?
2. What is the classification *rule or model* to best separate those groups?

Linear Discriminant Analysis (LDA)

For example, we want to know whether a soap product is good or bad based on several measurements on the product such as weight, volume, people's preferential score, smell, color contrast etc. The object here is soap. The class category or the group (💎good💎 and 💎bad💎) is what we are looking for (it is also called dependent variable). Each measurement on the product is called features that describe the object (it is also called independent variable).

Thus, in discriminant analysis, the dependent variable (Y) is the group and the independent variables (X) are the object features that might describe the group. The dependent variable is always category (nominal scale) variable while the independent variables can be any measurement scale (i.e. nominal, ordinal, interval or ratio).

If we can assume that the groups are linearly separable, we can use linear discriminant model (LDA). Linearly separable suggests that the groups can be separated by a linear combination of features that describe the objects. If only two features, the separators between objects group will become lines. If the features are three, the separator is a plane and the number of features (i.e. independent variables) is more than 3, the separators become a hyper-plane.

How does LDA work?

classification rule is to **assign an object to the group with highest conditional probability**. This is called Bayes Rule. This rule also minimizes the TEC. If there are g groups, the Bayes' rule is to assign the object to group i where $P(i | \mathbf{x}) > P(j | \mathbf{x})$, for $\forall j \neq i$.

We want to know the probability $P(i | \mathbf{x})$ that an object is belong to group i , given a set of measurement $\mathbf{x}$. In practice however, the quantity of $P(i | \mathbf{x})$ is difficult to obtain. What we can get is $P(\mathbf{x} | i)$. This is the probability of getting a particular set of measurement $\mathbf{x}$ given that the object comes from group i . For example, after we know that the soap is good or bad then we can measure the object (weight, smell, color etc.). What we want to know is to determine the group of the soap (good or bad) based on the measurement only.

Fortunately, there is a relationship between the two conditional probabilities that well known as Bayes Theorem:

$$P(i | \mathbf{x}) = \frac{P(\mathbf{x} | i) \cdot P(i)}{\sum_j P(\mathbf{x} | j) \cdot P(j)}$$

How does LDA work?

Thus, the Bayes' Rule becomes:

Assign the object to group i if

$$\frac{P(\mathbf{x}|i).P(i)}{\sum_{\forall k} P(\mathbf{x}|k).P(k)} > \frac{P(\mathbf{x}|j).P(j)}{\sum_{\forall k} P(\mathbf{x}|k).P(k)}, \text{ for } \forall j \neq i$$

The denominators for both sides of inequality are positive and the same, therefore we can cancel them out to become

Assign the object to group i if

$$P(\mathbf{x}|i).P(i) > P(\mathbf{x}|j).P(j), \text{ for } \forall j \neq i$$

How does LDA work?

If we have many classes and many dimension of measurement which each dimension will have many values, the computation of conditional probability $P(\mathbf{x}|i)$ requires a lot of data. It is more practical to assume that the data come from some theoretical distribution. The most widely used assumption is that our data come from Multivariate Normal distribution which formula is given as

$$P(\mathbf{x}|i) = \left(\frac{1}{(2\pi)^{\frac{n}{2}} |\mathbf{C}_i|^{\frac{1}{2}}} \right) \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^T \mathbf{C}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \right)$$

Where, $\boldsymbol{\mu}_i$ is vector mean and $\mathbf{C}_i$ is covariance matrix of group i . Inputting the distribution formula into Bayes rule we have:

Assign object with measurement $\mathbf{x}$ to group i if

$$\left(\frac{P(i)}{(2\pi)^{\frac{n}{2}} |\mathbf{C}_i|^{\frac{1}{2}}} \right) \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^T \mathbf{C}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \right) > \left(\frac{P(j)}{(2\pi)^{\frac{n}{2}} |\mathbf{C}_j|^{\frac{1}{2}}} \right) \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_j)^T \mathbf{C}_j^{-1} (\mathbf{x} - \boldsymbol{\mu}_j) \right), \quad i \neq j$$

Since factor of $(2\pi)^{\frac{n}{2}}$ are equal for both sides, we can cancel out

How does LDA work?

$$\left(\frac{P(i)}{|\mathbf{C}_i|^{\frac{1}{2}}} \right) \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)' \mathbf{C}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \right) > \left(\frac{P(j)}{|\mathbf{C}_j|^{\frac{1}{2}}} \right) \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_j)' \mathbf{C}_j^{-1} (\mathbf{x} - \boldsymbol{\mu}_j) \right), \quad i \neq j$$

Take logarithmic of both sides

$$-\frac{1}{2} \ln(|\mathbf{C}_i|) + \ln(P(i)) - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)' \mathbf{C}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) > -\frac{1}{2} \ln(|\mathbf{C}_j|) + \ln(P(j)) - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_j)' \mathbf{C}_j^{-1} (\mathbf{x} - \boldsymbol{\mu}_j), \quad i \neq j$$

Multiply both sides with -2, we need to change the sign of inequality

$$\ln(|\mathbf{C}_i|) - 2\ln(P(i)) + (\mathbf{x} - \boldsymbol{\mu}_i)' \mathbf{C}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) < \ln(|\mathbf{C}_j|) - 2\ln(P(j)) + (\mathbf{x} - \boldsymbol{\mu}_j)' \mathbf{C}_j^{-1} (\mathbf{x} - \boldsymbol{\mu}_j), \quad i \neq j$$

Deriving the LD Function

If all covariance matrices are equal $\mathbf{C} = \mathbf{C}_i = \mathbf{C}_j$, then we can simplify further into

$$\ln(|\mathbf{C}|) - 2\ln(P(i)) + (\mathbf{x} - \boldsymbol{\mu}_i)^T \mathbf{C}^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) < \ln(|\mathbf{C}|) - 2\ln(P(j)) + (\mathbf{x} - \boldsymbol{\mu}_j)^T \mathbf{C}^{-1} (\mathbf{x} - \boldsymbol{\mu}_j), \quad i \neq j$$

We can write $(\mathbf{x} - \boldsymbol{\mu}_i)^T \mathbf{C}^{-1} (\mathbf{x} - \boldsymbol{\mu}_i)$ into $\mathbf{x}\mathbf{C}^{-1}\mathbf{x}^T - 2\boldsymbol{\mu}_i\mathbf{C}^{-1}\mathbf{x}^T + \boldsymbol{\mu}_i\mathbf{C}^{-1}\boldsymbol{\mu}_i^T$. Thus, the inequality becomes

$$\ln(|\mathbf{C}|) - 2\ln(P(i)) + \mathbf{x}\mathbf{C}^{-1}\mathbf{x}^T - 2\boldsymbol{\mu}_i\mathbf{C}^{-1}\mathbf{x}^T + \boldsymbol{\mu}_i\mathbf{C}^{-1}\boldsymbol{\mu}_i^T < \ln(|\mathbf{C}|) - 2\ln(P(j)) + \mathbf{x}\mathbf{C}^{-1}\mathbf{x}^T - 2\boldsymbol{\mu}_j\mathbf{C}^{-1}\mathbf{x}^T + \boldsymbol{\mu}_j\mathbf{C}^{-1}\boldsymbol{\mu}_j^T, \quad i \neq j$$

We can cancel out the first and third terms (i.e. $\ln(|\mathbf{C}|)$ and $\mathbf{x}\mathbf{C}^{-1}\mathbf{x}^T$) of both sides because they do not affect the grouping decision. Thus, we have

$$-2\ln(P(i)) - 2\boldsymbol{\mu}_i\mathbf{C}^{-1}\mathbf{x}^T + \boldsymbol{\mu}_i\mathbf{C}^{-1}\boldsymbol{\mu}_i^T < -2\ln(P(j)) - 2\boldsymbol{\mu}_j\mathbf{C}^{-1}\mathbf{x}^T + \boldsymbol{\mu}_j\mathbf{C}^{-1}\boldsymbol{\mu}_j^T, \quad i \neq j$$

We multiply both sides of inequality with $-\frac{1}{2}$ (the sign of inequality reverse because we multiply with negative value), we have

$$\ln(P(i)) + \boldsymbol{\mu}_i\mathbf{C}^{-1}\mathbf{x}^T - \frac{1}{2}\boldsymbol{\mu}_i\mathbf{C}^{-1}\boldsymbol{\mu}_i^T > \ln(P(j)) + \boldsymbol{\mu}_j\mathbf{C}^{-1}\mathbf{x}^T - \frac{1}{2}\boldsymbol{\mu}_j\mathbf{C}^{-1}\boldsymbol{\mu}_j^T, \quad i \neq j$$

Let $f_i = \boldsymbol{\mu}_i\mathbf{C}^{-1}\mathbf{x}^T - \frac{1}{2}\boldsymbol{\mu}_i\mathbf{C}^{-1}\boldsymbol{\mu}_i^T + \ln(P(i))$, we have

Assign object with measurement $\mathbf{x}$ to group i if

$$f_i > f_j, \quad \text{for } \forall i \neq j$$

That is Linear Discriminant function

Linear Discriminant Function & Mahalanobis Distance

$$f_i = \mu_i \mathbf{C}^{-1} \mathbf{x}_k^T - \frac{1}{2} \mu_i \mathbf{C}^{-1} \mu_i^T + \ln(p_i)$$

Assign object k to group i that has maximum f_i

If you notice carefully the second term ($\mu_i \mathbf{C}^{-1} \mu_i^T$) is actually Mahalanobis distance, which is distance to measure dissimilarity between several groups.

Mahalanobis distance is also called *quadratic distance*. It measures the separation of two groups of objects. Suppose we have two groups with means $\bar{\mathbf{x}}_i$ and $\bar{\mathbf{x}}_j$, Mahalanobis distance is given by the following

Formula
$$d_{ij} = \left((\bar{\mathbf{x}}_i - \bar{\mathbf{x}}_j)^T \mathbf{S}^{-1} (\bar{\mathbf{x}}_i - \bar{\mathbf{x}}_j) \right)^{\frac{1}{2}}$$

The data of the two groups must have the same number of variables (the same number of columns) but not necessarily to have the same number of data (each group may have different number of rows).

Covariance matrix of group i is computed using centered data matrix $\hat{\mathbf{X}}$ The pooled covariance matrix of the two groups is computed as weighted average of the covariance matrices. The weighted average takes this form

$$\mathbf{C}_i = \frac{1}{n_i} \hat{\mathbf{X}}^T \hat{\mathbf{X}}$$

$$s_i(r, c) = \frac{1}{n} \sum_{i=1}^g n_i c_i(r, s)$$

Example

Factory ABC produces very expensive and high quality chip rings that their qualities are measured in term of curvature and diameter. Result of quality control by experts is given in the table below.

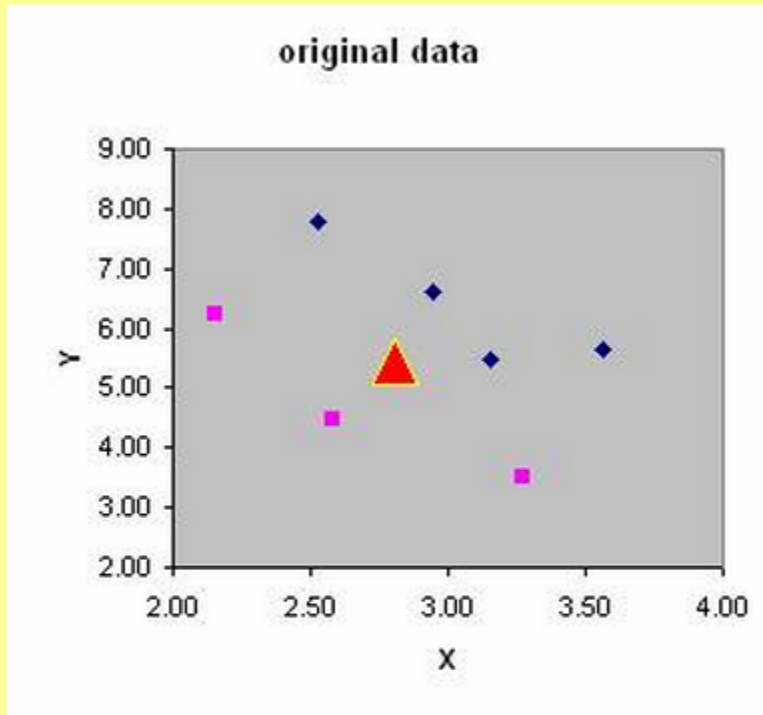
Curvature	Diameter	Quality Control Result
2.95	6.63	Passed
2.53	7.79	Passed
3.57	5.65	Passed
3.16	5.47	Passed
2.58	4.46	Not Passed
2.16	6.22	Not Passed
3.27	3.52	Not Passed

As a consultant to the factory, you get a task to set up the criteria for automatic quality control. Then, the manager of the factory also wants to test your criteria upon new type of chip rings that even the human experts are argued to each other. The new chip rings have curvature 2.81 and diameter 5.46.

Can you solve this problem by employing Discriminant Analysis?

Plotting Data

When we plot the features, we can see that the data is linearly separable. We can draw a line to separate the two groups. The problem is to find the line and to rotate the features in such a way to maximize the distance between groups and to minimize distance within group.



$\mathbf{x}$ = features (or independent variables) of all data. Each row (denoted by k) represents one object; each column stands for one feature.

$\mathbf{y}$ = group of the object (or dependent variable) of all data. Each row represents one object and it has only one column.

Example

$$\mathbf{x} = \begin{bmatrix} 2.95 & 6.63 \\ 2.53 & 7.79 \\ 3.57 & 5.65 \\ 3.16 & 5.47 \\ 2.58 & 4.46 \\ 2.16 & 6.22 \\ 3.27 & 3.52 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

In our example,

and

$\mathbf{x}_k$ = data of row k . For example, $\mathbf{x}_3 = [3.57 \ 5.65]$, $\mathbf{x}_7 = [3.27 \ 3.52]$

g = number of groups in $\mathbf{y}$. In our example, $g = 2$

$\mathbf{x}_i$ = features data for group i . Each row represents one object; each column stands for one feature. We separate $\mathbf{x}$ into several groups based on the number of category in $\mathbf{y}$.

$$\mathbf{x}_1 = \begin{bmatrix} 2.95 & 6.63 \\ 2.53 & 7.79 \\ 3.57 & 5.65 \\ 3.16 & 5.47 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 2.58 & 4.46 \\ 2.16 & 6.22 \\ 3.27 & 3.52 \end{bmatrix}$$

Example

μ_i = mean of features in group i , which is average of $\mathbf{x}_i$

$$\mu_1 = [3.05 \quad 6.38], \quad \mu_2 = [2.67 \quad 4.73]$$

μ = global mean vector, that is mean of the whole data set.

In this example, $\mu = [2.88 \quad 5.676]$

$\mathbf{x}_i^o$ = mean corrected data, that is the features data for group i , $\mathbf{x}_i$, minus the global mean vector μ

$$\mathbf{x}_1^o = \begin{bmatrix} 0.060 & 0.951 \\ -0.357 & 2.109 \\ 0.679 & -0.025 \\ 0.269 & -0.209 \end{bmatrix}, \quad \mathbf{x}_2^o = \begin{bmatrix} -0.305 & -1.218 \\ -0.732 & 0.547 \\ 0.386 & -2.155 \end{bmatrix}$$

$$\mathbf{c}_i = \frac{(\mathbf{x}_i^o)^T \mathbf{x}_i^o}{n_i} = \text{covariance matrix of group } i$$

$$\mathbf{c}_1 = \begin{bmatrix} 0.166 & -0.192 \\ -0.192 & 1.349 \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} 0.259 & -0.286 \\ -0.286 & 2.142 \end{bmatrix}$$

Example

$$C(r, s) = \frac{1}{n} \sum_{i=1}^g n_i \cdot c_i(r, s)$$
 = pooled within group covariance matrix. It is calculated for each entry (r, s) in the matrix. In our example, $\frac{4}{7} \cdot 0.166 + \frac{3}{7} \cdot 0.259 = 0.206$,
 $\frac{4}{7}(-0.192) + \frac{3}{7}(-0.286) = -0.233$ and $\frac{4}{7} \cdot 1.349 + \frac{3}{7} \cdot 2.142 = 1.689$, therefore

$$\mathbf{C} = \begin{bmatrix} 0.206 & -0.233 \\ -0.233 & 1.689 \end{bmatrix}$$

$$\mathbf{C}^{-1} = \begin{bmatrix} 5.745 & 0.791 \\ 0.791 & 0.701 \end{bmatrix}$$

The inverse of the pooled covariance matrix is

$\mathbf{p}$ = prior probability vector (each row represent prior probability of group i). If we do not know the prior probability, we just assume it is equal to total sample of each group divided by the

total samples, that is $p_i = \frac{n_i}{N}$

$$\mathbf{p} = \begin{bmatrix} 0.571 \\ 0.429 \end{bmatrix} = \begin{bmatrix} \frac{4}{7} \\ \frac{3}{7} \end{bmatrix}$$

Example

Discriminant function

$$f_i = \mu_i \mathbf{C}^{-1} \mathbf{x}_k^T - \frac{1}{2} \mu_i \mathbf{C}^{-1} \mu_i^T + \ln(p_i)$$

We should assign object k to group i that has maximum f_i

The results of our computation are given in MS Excel as shown in the figure below.

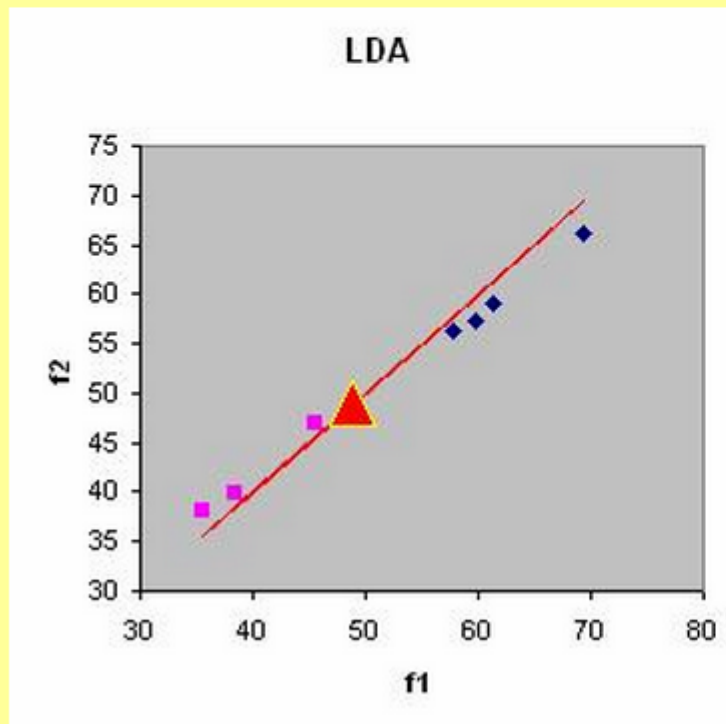
	A	B	C	D	E	F	G	H	I	J	K
1	Linear Discriminant Analysis: 2 Category 2 Dimensional Data										
2											
3	Training Data, D			Mean Corrected Data		Discriminant function		Results			
4	class	X1	X2	X1o	X2o		f1	f2	Classification		
5	1	2.95	6.63	0.060	0.951		55.220	53.071			1
6	1	2.53	7.79	-0.357	2.109		53.774	51.394			1
7	1	3.57	5.65	0.679	-0.025		62.476	59.589			1
8	1	3.16	5.47	0.269	-0.209		51.953	50.764			1
9	2	2.58	4.46	-0.305	-1.218		32.028	34.313			2
10	2	2.16	6.22	-0.732	0.547		34.554	35.757			2
11	2	3.27	3.52	0.386	-2.155		41.174	42.414			2
12	prediction	2.81	5.46	-0.078	-0.219		44.049	44.085			2
13											
14	average	2.888	5.676								
15	std dev	0.454	1.300								

The discriminant function is our classification rules to assign the object into separate group. If we input the new chip rings that have curvature 2.81 and diameter 5.46, reveal that it does not pass the quality control.

Graphing the Points

The discriminant function is our classification rules to assign the object into separate group. If we input the new chip rings that have curvature 2.81 and diameter 5.46, reveal that it does not pass the quality control.

Transforming all data into discriminant function (f_1, f_2) we can draw the training data and the prediction data into new coordinate. The discriminant line is all data of discriminant function $X = \{f_1, f_2\}$ and $Y = \{f_1, f_2\}$. In MS Excel, you can hold CTRL key while dragging the second region to select both regions.



Things to do

- Read LDA here:

<http://people.revoledu.com/kardi/tutorial/LDA/LDA.html>

Acknowledgements for material:

Teknomo, K. (2015). Discriminant Analysis Tutorial.

<http://people.revoledu.com/kardi/tutorial/LDA/LDA.html>



Questions or Comments?